

Hierarchical Inference in Gaussian Models

As we saw in the preceding sections, the *parameters* $\Theta := \{m, k, A, b, \Lambda\}$ of a Gaussian model,

$$p(f | \Theta) = \mathcal{N}(f; m, k), \quad p(y | f, \Theta) = \mathcal{N}(y; Af + b, \Lambda),$$

fundamentally affect the shape of the posterior on f . So what should be done if the right choice of parameters is itself unknown? The difference between a *variable* (a number of interest) and a *parameter* (a nuisance number required to specify the model, but not of central interest) is only conceptual, not formal. If the right value for a parameter is not clear a priori, the natural treatment is to assign uncertainty to it, too. For this purpose, the evidence term

$$p(y | \Theta) = \int p(f | \Theta) p(y | f, \Theta) df = \mathcal{N}(y; Am + b, k + \Lambda)$$

provides a *marginal likelihood* (also known as the *type-II likelihood*) for the parameters.¹

Because inferring parameters involves “modelling the model”, this notion is known as *hierarchical inference*. Hierarchical inference can increase the computational cost of the overall estimation, so it is not always advisable. But for some special cases of general interest, analytical forms are available. Usually, the probability distribution over the parameters itself contains numbers specifying certain parts. Those numbers are called *hyperparameters*. If one truly wanted, they too could be inferred via additional hierarchical modelling layers, *ad infinitum*, but this is not done in practice to keep computational cost finite. We will now outline such a framework for estimating the unknown mean and variance of a Gaussian model. We will employ a *conjugate prior*² over these hyperparameters, i.e. a prior which

¹ More on the kind of exact marginalisation performed below, with extensions, can be found in a book by Bretthorst (1988). A more compact discussion of conjugate priors for the Gaussian is in §24.1 of MacKay (2003).

² An early work on conjugate priors is by Edwin Pitman (1936). An general exposition of conjugate priors for exponential families was provided by Diaconis and Ylvisaker (1979). The core idea behind conjugate priors, namely to formally treat prior knowledge like additional data, is arguably first used, albeit without much discussion, in §VI of Pierre-Simon Laplace’s *Théorie analytique des probabilités* (1814, p. 364), in the famous argument about inferring probabilities from observed events, which also introduces the Gaussian approximation to the Beta integral now known as the *Laplace approximation* (1814, p. 365). There, Laplace argues that one can always use a uniform prior, since non-uniform prior knowledge can be represented by adding counts to the observations.

ensures that the posterior distribution over these hyperparameters, given data, has the same (Gaussian) form.

► 6.1 Inference on Scalar Parameters of Gaussians

Assume the real numbers $\mathbf{y} := [y_i]_{i=1,\dots,N}$ are drawn i.i.d. from a Normal distribution with unknown mean α and variance β^2 ,

$$p(\mathbf{y} \mid \alpha, \beta) = \prod_{i=1}^N \mathcal{N}(y_i; \alpha, \beta^2) = \mathcal{N}(\mathbf{y}; \alpha \mathbf{1}, \beta^2 I).$$

We would like to assign probability densities over the latent quantities α and β . Purely from an algebraic perspective (as opposed to a deeper philosophical motivation for this kind of uncertainty), it is convenient to choose a prior distribution $p(\alpha, \beta)$ of some form $p(\alpha, \beta) = \eta(\alpha, \beta; \theta)$ with hyperparameter θ , such that the posterior distribution has the same functional form η as the prior, just with updated hyperparameters $\tilde{\theta}$ instead of θ :

$$p(\alpha, \beta \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \alpha, \beta) \eta(\alpha, \beta; \theta)}{\int p(\mathbf{y} \mid \alpha, \beta) \eta(\alpha, \beta; \theta) d\alpha d\beta} = \eta(\alpha, \beta; \tilde{\theta}).$$

A prior η with this property is called *conjugate* to the likelihood $p(\mathbf{y} \mid \alpha, \beta)$. For the present case of a Gaussian likelihood with latent mean and variance, the standard choice of conjugate prior assigns a Gaussian distribution of mean μ_0 and variance proportional to β^2 (with scale λ_0) to α , and a Gamma distribution with parameters a_0, b_0 to the *inverse* of β . This is known as the *Gauss-Gamma*, or *Gauss-inverse-Gamma* prior, and has the hyperparameters $\theta_0 := (\mu_0 \quad \lambda_0 \quad a_0 \quad b_0)$:

$$\begin{aligned} \eta(\alpha, \beta \mid \mu_0, \lambda_0, a_0, b_0) &= p(\alpha \mid \beta, \mu_0, \lambda_0) p(\beta \mid a_0, b_0) \\ &= \mathcal{N}\left(\alpha; \mu_0, \frac{\beta^2}{\lambda_0}\right) \mathcal{G}(\beta^{-2}; a_0, b_0), \end{aligned}$$

$$\text{where } \mathcal{G}(z; a, b) := \frac{b^a z^{a-1}}{\Gamma(a)} e^{-bz}.$$

Here, $\mathcal{G}(\cdot; a, b)$ is the Gamma distribution with shape $a > 0$ and rate $b > 0$. The normalisation constant of the Gamma distribution, and also the source of the distribution's name, is the Gamma function Γ .³ To compute the posterior, we multiply prior and likelihood, and identify

$$\begin{aligned} p(\alpha, \beta \mid \mathbf{y}) &\propto p(\mathbf{y} \mid \alpha, \beta) p(\alpha, \beta) \\ &= \mathcal{N}(\mathbf{y}; \mathbf{1}\alpha, \beta^2 I) \mathcal{N}(\alpha; \mu_0, \beta^2/\lambda_0) \mathcal{G}(\beta^{-2}; a_0, b_0). \end{aligned} \quad (6.2)$$

³ The Gamma function is an extension of the factorial function. The Greek symbol for this function is due to Legendre who utilised two “Eulerian integrals”, B, Γ (the first symbol being a capital Greek letter Beta):

$$\begin{aligned} B(m, n) &= \int_0^1 x^{m-1} (1-x)^{n-1} dx, \\ \Gamma(t) &= \int_0^1 e^{-x} x^{t-1} dx. \end{aligned} \quad (6.1)$$

In the context of this text, it is a neat marginal observation that, while Legendre was interested in problems of chance, Euler's original motivation for considering those integrals, in an exchange with Goldbach, was one of *interpolation*: he was trying to find a “simple” smooth function connecting the factorial function on the reals. And indeed,

$$\Gamma(n) = (n-1)! \quad \forall n \in \mathbb{N}_{\setminus 0}.$$

Legendre is also to blame for this unsightly shift in the function's argument, since he constructed Eq. (6.1) by rearranging Euler's more direct result

$$n! = \int_0^1 (-\log x)^n dx.$$

A great exposition on this story and the Gamma function can be found in a Chauvenet-prize-decorated article by Davis (1959). It is left as a research exercise to the reader to consider in which sense Euler's answer to this interpolation problem is natural, in particular from a probabilistic-numerics standpoint (that is, which prior assumptions give rise to the Gamma function as an interpolant of the factorials, whether there are other meaningful priors yielding other interpolations).

We first deal with the Gaussian part, using Eq. (3.5) (and some simple vector arithmetic) to re-arrange this expression as

$$\mathcal{G}(\beta^{-2}; a_0, b_0) \mathcal{N}\left(\mathbf{y}; \mu_0, \beta^2 \left(I + \frac{\mathbf{1} \cdot \mathbf{1}^\top}{\lambda_0}\right)\right) \cdot \mathcal{N}\left(\alpha; \frac{\lambda_0 \mu_0 + \sum_i y_i}{\lambda_0 + N}, \frac{\beta^2}{\lambda_0 + N}\right). \quad (6.3)$$

The second Gaussian expression is evidently a Gaussian over α , and the first one does not depend on α . To deal with the first part, we use the matrix inversion lemma, Eq. (15.9), to rewrite

$$\left(I + \frac{\mathbf{1} \cdot \mathbf{1}^\top}{\lambda_0}\right)^{-1} = I - \frac{\mathbf{1} \cdot \mathbf{1}^\top}{N + \lambda_0}.$$

This allows writing the first line of Eq. (6.3) more explicitly (leaving out normalisation constants independent of β) as

$$\begin{aligned} &\mathcal{G}(\beta^{-2}; a_0, b_0) \cdot \mathcal{N}\left(\mathbf{y}; \mu_0, \beta^2 \left(I + \frac{\mathbf{1} \cdot \mathbf{1}^\top}{\lambda_0}\right)\right) \\ &\propto \left(\frac{1}{\beta^2}\right)^{a_0-1} \exp\left(-\frac{b_0}{\beta^2}\right) \cdot \left(\frac{1}{\beta^2}\right)^{N/2} \cdot \exp\left(-\frac{1}{2\beta^2} \left(\sum_{i=1}^N (y_i - \mu_0)^2 - \frac{(\sum_{i=1}^N y_i - N\mu_0)^2}{\lambda_0 + N}\right)\right). \end{aligned} \quad (6.4)$$

At this point, it is common practice to introduce the following substitutions, known as *sufficient statistics*:

$$\bar{\alpha} = \frac{1}{N} \sum_{i=1}^N y_i \quad \text{sample mean,} \quad (6.5)$$

$$\bar{\beta}^2 = \frac{1}{N} \sum_{i=1}^N (y_i - \bar{\alpha})^2 \quad \text{sample variance,} \quad (6.6)$$

with which the cumbersome expression in Eq. (6.4) simplifies (by shuffling sums around) to

$$\begin{aligned} &\sum_{i=1}^N (y_i - \mu_0)^2 - \frac{(\sum_{i=1}^N y_i - N\mu_0)^2}{\lambda_0 + N} \\ &= N\bar{\beta}^2 + \frac{\lambda_0 N}{\lambda_0 + N} (\bar{\alpha} - \mu_0)^2. \end{aligned}$$

The terms in Eq. (6.4) thus have the form of a Gamma distribution over β^{-2} . All other terms suppressed by the $\propto$ sign make up the normalisation constant of the new Gauss-Gamma-posterior

$$p(\alpha, \beta \mid \mathbf{y}, \mu_0, \lambda_0, a_0, b_0) = \eta(\alpha, \beta \mid \mu_N, \lambda_N, a_N, b_N), \quad (6.7)$$

with the new parameters

$$\begin{aligned} \mu_N &= \frac{\lambda_0 \mu_0 + N\bar{\alpha}}{\lambda_0 + N}, \quad \lambda_N = \lambda_0 + N, \\ a_N &= a_0 + N/2, \quad b_N = b_0 + 1/2 \left(N\bar{\beta}^2 + \frac{\lambda_0 N}{\lambda_0 + N} (\bar{\alpha} - \mu_0)^2 \right). \end{aligned} \quad (6.8)$$

Exercise 6.1 (easy). Assume that, instead of i.i.d. draws y_i , one observes individually scaled samples $a_i = y_i s_i$ with known scales s_i . By re-tracing the derivation, convince yourself that this situation can be addressed in the same fashion, but with new sufficient statistics

$$\begin{aligned} \bar{\alpha} &= \frac{1}{N} \sum_{i=1}^N \frac{a_i}{s_i}, \\ \bar{\beta}^2 &= \frac{1}{N} \sum_{i=1}^N \left(\frac{a_i}{s_i} - \bar{\alpha} \right)^2. \end{aligned}$$

This will be helpful for hierarchical inference in §III.

The common interpretation for this result is that λ_N and a_N can be seen as *counters* for the number of observations collected, while μ_N and b_N are sufficient statistics for the population's mean and variance.⁴ In this sense, λ_0 and $2a_0$ amount to pseudo-observations (see note 2) encoded in the prior. As N increases, μ_N (the expected value of α under the Gauss-Gamma prior) converges to the sample mean $\bar{\alpha}$, while $\mathbb{E}(\beta^2) = b_N/a_N$ converges to the sum of the sample variance and a correction term

$$\frac{\lambda_0 N}{\lambda_0 + N} (\bar{\alpha} - \mu_0)^2.$$

Intuitively, this term corrects for the fact that $\bar{\beta}^2$ is a biased estimate of the variance – the sample mean is typically closer to the samples than the actual mean is, and this bias depends on how far the initial estimate μ_0 is from the correct mean.

In applications our main interest will usually not be in the posterior distribution over the parameters (α, β) , but predictions of the subsequent sample y_{N+1} . Under the hierarchical Gauss-Gamma model, this posterior distribution is the *marginal* distribution reached by integrating out the *posterior* over the hyperparameters,

$$p(y_{N+1} | \mathbf{y}) = \int_{-\infty}^{\infty} \int_0^{\infty} p(y_{N+1} | \mathbf{y}, \alpha, \beta) p(\alpha, \beta | \mathbf{y}) d\alpha d\beta.$$

The integral over α is easy, as it is an instance of the basic Gaussian properties above. The remaining integral over β is over the product of a Gaussian and several Gamma terms, which is given by a famous result known as⁵ *Student's-t* distribution:

$$\begin{aligned} \text{St}(y_{N+1}; \mu_N, a_N/b_N, 2a_N) \\ &= \int_0^{\infty} \mathcal{N}(y_{N+1}; \mu_N, \beta^2) \mathcal{G}(\beta^{-2}; a_N, b_N) d\beta \quad (6.9) \\ &= \frac{\Gamma(a_N + 1/2)}{\Gamma(a_N)} \frac{b_N^{a_N}}{\sqrt{2\pi}} \left(b_N + \frac{(y_{N+1} - \mu_N)^2}{2} \right)^{-a_N - 1/2}. \end{aligned}$$

► 6.2 Inference on Multivariate Gaussian Parameters

The derivations above can be extended to the case of *multivariate* observations $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_N] \in \mathbb{R}^{M \times N}$ drawn i.i.d. as

$$p(\mathbf{Y} | \boldsymbol{\alpha}, B) = \prod_i \mathcal{N}(\mathbf{y}_i; \boldsymbol{\alpha}, B) \quad (6.10)$$

from a Gaussian with unknown mean vector $\boldsymbol{\alpha}$ and unknown covariance matrix B . The corresponding conjugate prior is the

⁴ The expected value of a Gamma-distributed random variable is

$$\mathbb{E}_{\mathcal{G}(z; a, b)}(z) = \frac{a}{b}.$$

⁵ An alternative contender for the invention of this distribution is the German geodesist F.R. Helmert, but the endearing story of the statistician Gosset, writing under the pseudonym “Student” (1908) so as not to violate a non-disclosure agreement with the Guinness Brewery, his employer, has stuck. In fact, Helmert’s 1875 letter to the *Zeitschrift für Mathematik und Physik* is an entertaining read, too. It contains a dressing-down of R. A. Mees (Professor of Physics at Göttingen) who, in an article published in the same newspaper some months prior (pp. 145 ff.), apparently misread a discussion, by Helmert, of Gauss’ work on estimation of errors. The letter opens with the sentence “The point of the following is to show that Mr. Mees errs, not just in his evaluation of my work, but throughout his entire essay.”

Gauss-inverse-Wishart distribution

$$p(\boldsymbol{\alpha}, B) = \mathcal{N}(\boldsymbol{\alpha}; \boldsymbol{\mu}_0, 1/\lambda_0 B) \mathcal{W}(B^{-1}; W_0, \nu_0), \quad (6.11)$$

with the parameters $\boldsymbol{\mu}_0 \in \mathbb{R}^M$, $\lambda_0 \in \mathbb{R}_+$, a symmetric positive definite matrix W_0 , and the so-called degree of freedom $\nu_0 > M - 1$. Here, $\mathcal{W}(B^{-1}; W_0, \nu_0)$ is a *Wishart distribution*,⁶

⁶ Wishart (1928)

$$\mathcal{W}(B^{-1}; W_0, \nu_0) := \frac{|B^{-1}|^{(\nu_0 - M - 1)/2} \exp\left(-1/2 \operatorname{tr}(W_0^{-1} B^{-1})\right)}{2^{\nu_0 M/2} |W_0|^{\nu_0/2} \Gamma_M(\nu_0/2)},$$

using what is sometimes called the *multivariate Gamma function*,

$$\Gamma_M(x) := \pi^{M(M-1)/4} \prod_{j=1}^M \Gamma\left(x + \frac{(1-j)}{2}\right).$$

The posterior resulting from the likelihood (6.10) and prior (6.11) emerges analogously to the derivations above. It is a Gauss-inverse-Wishart with the updated parameters

$$\begin{aligned} \boldsymbol{\mu}_N &= \frac{\lambda_0 \boldsymbol{\mu}_0 + N \bar{\boldsymbol{\alpha}}}{\lambda_0 + N}, \\ \lambda_N &= \lambda_0 + N, \\ \nu_N &= \nu_0 + N, \\ W_N &= W_0 + N \bar{B} + \frac{\lambda_0 N}{\lambda_0 + N} (\bar{\boldsymbol{\alpha}} - \boldsymbol{\mu}_0)(\bar{\boldsymbol{\alpha}} - \boldsymbol{\mu}_0)^\top, \end{aligned}$$

using the multivariate forms of the sufficient statistics from Eqs. (6.5) and (6.6):

$$\begin{aligned} \bar{\boldsymbol{\alpha}} &= \frac{1}{N} \sum_i \mathbf{y}_i, \\ \bar{B} &= \frac{1}{N} \sum_i (\mathbf{y}_i - \bar{\boldsymbol{\alpha}})(\mathbf{y}_i - \bar{\boldsymbol{\alpha}})^\top. \end{aligned}$$

⁷ When comparing Eq. (6.12) to Eq. (6.9), an unexpected factor of 2 shows up here and there. This comes from differing definitions of the “counters” a_N and ν_N , which in turn are due to the standard definition of the Gamma distribution.

As above, it is also possible to marginalise over this multivariate posterior. This again yields a Student- t posterior, but a multivariate one:⁷

$$\begin{aligned} p(\mathbf{y}_{N+1} \mid \mathbf{Y}, \mu_0, \lambda_0, \nu_0, W_0) &= \int p(\mathbf{y}_{N+1} \mid \boldsymbol{\alpha}, B) p(\boldsymbol{\alpha}, B \mid \mathbf{Y}, \mu_0, \lambda_0, \nu_0, W_0) d\boldsymbol{\alpha} dB \\ &= \operatorname{St}_M\left(\mathbf{y}_{N+1}; \mu_N, \frac{\lambda_N + 1}{\lambda_N(\nu_N - M + 1)} W_N, \nu_N - M + 1\right) \\ &= \frac{\Gamma(\nu_N + M/2)}{\Gamma(\frac{\nu_N}{2}) \nu_N^{M/2} \pi^{M/2} |W_N|^{1/2}} \left[1 + \frac{1}{\nu_N} (\mathbf{y}_{N+1} - \mu_N)^\top W_N^{-1} (\mathbf{y}_{N+1} - \mu_N)\right]^{-(\nu_N + M)/2}. \end{aligned} \quad (6.12)$$

► 6.3 Conjugate Prior Hierarchical Inference for Filters

For linear Gaussian state-space models as defined by Eqs. (5.8)–(5.9), the marginal likelihood factorises over the Markov chain:⁸

$$p(\mathbf{y} \mid \theta) = \prod_{i=1}^N p(y_i \mid \mathbf{y}_{1:i-1}, \theta) = \prod_{i=1}^N \mathcal{N}(y_i; Hm_i^-, HP_i^- H^\top). \quad (6.13)$$

In particular, for SDEs like those of the preceding §5.4 and §5.5, which can be written, with an explicit role for θ , as

$$dz(t) = Fz(t) dt + \theta \tilde{L} d\omega_t,$$

hierarchical inference can be formulated efficiently in the filter formulation:⁹ define the unit-scale predictive variance analogously to Eq. (5.21) as

$$\tilde{Q}_{t_i} := \int_0^{t_{i+1}-t_i} e^{F\tau} \tilde{L} \tilde{L}^\top e^{F\tau} d\tau,$$

which extracts θ in Eq. (6.13). We get

$$p(\mathbf{y} \mid \theta) = \prod_{i=1}^N \mathcal{N}(y_i; Hm_i^-, \theta^2 H \tilde{P}_i^- H^\top)$$

with a recursively defined $\tilde{P}^- =: A \tilde{P} A^\top + \tilde{Q}$, started at $\tilde{P}_0^- = P_0^-$. Given the inverse Gamma prior

$$p(\theta) = \mathcal{G}(\theta^{-2}, \alpha_0, \beta_0),$$

the posterior on θ can then be updated recursively during the filtering pass, and the analogue to Eq. (6.7) simplifies to

$$\begin{aligned} p(\theta^{-2} \mid \mathbf{y}_{1:N}) &= \mathcal{G}\left(\theta^{-2}; \alpha_0 + \frac{N}{2}, \beta_0 + \frac{1}{2} \sum_{i=1}^N \frac{(y_i - Hm_i^-)^2}{H \tilde{P}_i^- H^\top}\right) \\ &=: \mathcal{G}(\theta^{-2}, \alpha_N, \beta_N). \end{aligned} \quad (6.14)$$

In Algorithm 5.1, this can be realised¹⁰ by using $\tilde{Q}$ instead of Q , and adding the lines $\alpha \leftarrow \alpha + 1/2$ and $\beta \leftarrow \beta + z/2s$ after line 8 (those two variables having been initialised to $\alpha \leftarrow \alpha_0, \beta \leftarrow \beta_0$). The corresponding Student- t marginal on the state can be computed locally, if necessary, in each step.

⁸ This is known as the *prediction error decomposition*, See Eq. (12.5) in Särkkä (2013).

⁹ As in the preceding sections, analytic hierarchical inference on the scale θ requires noise-free observations. The same applies here. In the notation of the Kalman filter, this amounts to $R = 0$.

¹⁰ Pseudo-code can be found in the following chapter, as Algorithm 11.2.